

The Program in Action

When you run *fibonacci.py*, the output will look like this:

```
Fibonacci Sequence, by Al Sweigart al@inventwithpython.com
--snip--
Enter the Nth Fibonacci number you wish to
calculate (such as 5, 50, 1000, 9999), or QUIT to quit:
> 50

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584,
4181, 6765, 10946, 17711, 28657, 46368, 75025, 121393, 196418, 317811, 514229,
832040, 1346269, 2178309, 3524578, 5702887, 9227465, 14930352, 24157817,
39088169, 63245986, 102334155, 165580141, 267914296, 433494437, 701408733,
1134903170, 1836311903, 2971215073, 4807526976, 7778742049
```

How It Works

Because Fibonacci numbers quickly become very large, lines 46 to 50 check if the user has entered a number that's 10,000 or larger and displays a warning that it may take some time for the output to finish displaying on the screen. While your programs can quickly do millions of calculations almost instantly, printing text to the screen is relatively slow and could take several seconds. The warning in our program reminds the user they can always terminate the program by pressing CTRL-C.

```
1. """Fibonacci Sequence, by Al Sweigart al@inventwithpython.com
2. Calculates numbers of the Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13...
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: short, math"""
5.
6. import sys
7.
8. print('''Fibonacci Sequence, by Al Sweigart al@inventwithpython.com
9.
10. The Fibonacci sequence begins with 0 and 1, and the next number is the
11. sum of the previous two numbers. The sequence continues forever:
12.
13. 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987...
14. ''')
15.
16. while True: # Main program loop.
17.     while True: # Keep asking until the user enters valid input.
18.         print('Enter the Nth Fibonacci number you wish to')
19.         print('calculate (such as 5, 50, 1000, 9999), or QUIT to quit:')
20.         response = input('> ').upper()
21.
22.         if response == 'QUIT':
23.             print('Thanks for playing!')
24.             sys.exit()
25.
26.         if response.isdecimal() and int(response) != 0:
27.             nth = int(response)
```